

VISORIA: AN ATTENTION-AWARE INTERVENTION SYSTEM FOR ONLINE CLASSROOMS

by

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This project titled, “**Visoria: An Attention-Aware Intervention System for Online Classrooms**”, submitted by Mostafa Shariare, Omor Faruk, Roll No.: 2202045, 2202026, Session: 2022–23, has been accepted as satisfactory in partial fulfillment of the requirement for the degree of Bachelor of Science in Educational Technology and Engineering on September 2026.

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Candidate's Declaration

This is to certify that the work presented in this project entitled, “Visoria: An Attention-Aware Intervention System for Online Classrooms”, is the outcome of the research carried out by Mostafa Shariare, Omor Faruk under the supervision of Farhana Islam, Assistant Professor, Department of Educational Technology and Engineering, University of Frontier Technology, Bangladesh.

It is also declared that neither this project nor any part thereof has been submitted anywhere else for the award of any degree, diploma, or other qualifications.

Signature of the Candidate

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Dedication

*To our beloved parents, whose unconditional love, endless sacrifices,
and continuous prayers have illuminated every step of our lives;*

*To our respected teachers and mentors, whose wisdom, guidance, and
encouragement have shaped our academic journey;*

*And to dedicated educators everywhere, who strive to inspire, innovate,
and transform the future of learning.*

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Mostafa Shariare
Omor Faruk

Abstract

Student attention is a critical predictor of learning outcomes, yet it remains challenging to observe in online learning environments where teachers lack access to traditional nonverbal cues. To address this issue, we propose Visoria, an attention-aware intervention system designed for online classrooms. Visoria utilizes an edge-based, privacy-preserving computer vision pipeline that processes webcam feeds locally to extract behavioral features. We employ Mediapipe for facial landmarks and 3D pose estimation, and YOLOv8n for object detection, capturing 19 distinct behavioral features. These features are classified using a Random Forest model, followed by temporal smoothing over a 15-frame window and a rule-based score fusion mechanism (55% machine learning, 45% rule-based) to robustly predict student attention states. Experimental results on a test set of 800 samples demonstrate that our Random Forest model achieves an exceptional accuracy of 99.88%, with a precision of 100% and an F1-score of 99.85%. Furthermore, SHAP (SHapley Additive exPlanations) analysis highlights forward head pose as the most significant predictor of attention. Beyond classification, Visoria introduces a comprehensive end-to-end platform featuring a Teacher Management Portal and a Student Web Portal. This enables instructors to monitor live classroom engagement and dispatch interactive interventions, such as pop quizzes and reflection prompts, to effectively re-engage distracted learners. Ultimately, Visoria bridges the engagement gap in digital education by combining accurate, privacy-first attention tracking with actionable teaching tools.

Chapter 1

Introduction

1.1 Problem Statement

Student attention is one of the strongest predictors of learning outcomes, but can be much harder to observe in an online setting. In a physical classroom, a teacher can read the room, picking up on nonverbal queues like wandering eyes or slouched posture and adjusting based on the classroom dynamic. In online and blended learning spaces, the teacher has very little opportunity to perform a “room-reading strategy”, often seeing only the small videos of their students, or no video at all.

A number of works explore ways to bridge this divide using computer vision. Camera-based systems have been designed to detect student behaviors and emotions using object detectors such as YOLOv5 combined with tracking algorithms [1], to estimate where a student’s gaze is directed using facial landmark and 3D pose analysis [2], and to classify attention based on webcam features such as head pose, hand position, and phone presence using classical ML classifiers [3]. Public benchmark datasets, such as DAiSEE, have allowed for an alternative line of work that investigates engagement as a temporal, video-level property rather than a single-frame classification [4–6].

However, this project identifies several unresolved gaps across the existing literature that motivate the present work:

Deployability versus accuracy. The most effective attention-detection systems currently utilize specialized hardware such as commercial eye trackers [7,8] or multi-track deep learning pipelines that require hefty amounts of computation [9,10], limiting them to controlled or well-endowed environments. Systems that utilize more common and affordable webcam sensors [3,10–12] are much more widely deployable, and have not been evaluated consistently across a range of classroom environments and camera hard-

ware [1, 3, 11].

Privacy and bandwidth. Some systems require that video imagery or rich visual data be sent to a central server for processing and analysis, which can be problematic from both a privacy (e.g., FERPA, GDPR) and bandwidth perspective. A number of proposals have been made for moving much of this processing out to the edge to avoid these issues, while still allowing higher level analysis of the video content at the server [3, 9, 11]. In some cases, the proposed solutions do not adequately address privacy [11].

Detection without action. The vast majority of the surveyed work considers stopping or scoring attention/engagement [1–6, 8–16], leaving action for another time. Only one study in this survey explicitly measured the effect of a suggested action due to inattention to confirm if it led to improved student engagement [7]. However, this work used special eye-tracking hardware rather than a standard webcam, limiting applicability.

The offline-to-real-time accuracy gap. Detection accuracy measured offline, on pre-recorded data, is not necessarily reflective of performance on a live, deployed system. One work found that real-time, classroom, detection accuracy ($F1 = 0.40$) was significantly lower than offline accuracy on the same task ($F1 = 0.59$) [8], a discrepancy rarely reported or measured in other reviewed works.

Frame-level versus temporal modeling. Several works claim that engagement is a time dependent property of the learning session and not a momentary state [4, 5], but the majority of deployed, webcam based systems operate on independent frame by frame classifications [1–3, 10–13] having no mechanism to regularize these noisy varying estimations into a stable state over time.

In summary, none of the surveyed systems have been able to offer simultaneous standard-webcam deployment, preserve privacy at the device level where raw video is not transmitted, maintain stable temporal attention estimates, provide explainability, and employ a structured, teacher-mediated intervention mechanism, all of which are addressed by this project.

1.2 Objectives of the Project

The objectives of this project are to:

1. Design and implement a privacy-preserving, edge-based student attention monitoring system (Visoria) that uses an off-the-shelf consumer webcam, eliminating the need for specialized hardware while keeping video analysis local to mitigate

bandwidth and privacy concerns [3, 7–9].

2. Develop a robust attention classification model that fuses rule-based scoring with machine learning predictions, and apply temporal smoothing to create a continuous, stable estimate of the student’s attention state over time [1–3, 9, 11, 13–15].
3. Integrate explainable AI (XAI) and actionable intervention mechanisms to provide human-comprehensible justifications for attention predictions, enabling teachers to trigger guided interventions when student attention drops [7, 14].
4. Evaluate the system’s overall performance, measuring its classification accuracy, robustness to different conditions, and explainability, while identifying limitations and the boundaries of its generalizability across diverse educational settings [1, 5, 11, 13].

1.3 Project Outline

The rest of this project is structured as follows. **Chapter 2** presents the background and related work about computer-vision-based attention and engagement monitoring, privacy-preserving edge architectures, and attention-aware intervention systems, all of which are placed in the context of this project to identify the gaps addressed by it. **Chapter 3** describes the methodology of the Visoria system, including its system architecture, facial and behavioral feature extraction pipeline, machine learning and rule-based attention estimation approach, temporal smoothing, explainability layer, and teacher alert and guided intervention mechanisms. **Chapter 4** presents the experimental results and evaluation of the system’s trained classifier, including its accuracy, confusion matrix, ROC-AUC, SHAP-based explainability analysis, and deployed system user interfaces. Finally, **Chapter 5** concludes the project by summarizing the results, discussing the limitations of the present work, and suggesting future research directions.

Chapter 2

Literature Review

Student attention and engagement monitoring has become an important research area in educational technology, particularly with the increasing use of computer vision, machine learning, and multimodal sensing in classroom and online learning environments. Existing studies have explored different approaches, ranging from facial and gaze-based analysis to multimodal behavioral features and edge-based intelligence. The following studies provide an overview of the development of automated student attention and engagement monitoring systems.

2.1 Computer Vision-Based Student Attention Monitoring

Canedo et al. [11] proposed a system for determining the degree to which students are engaged and identifying behaviors that suggest engagement falls below a certain threshold through the use of an ordinary camera. The authors first examined the state of the art in the field of computer vision as it applies to face detection, face recognition, and behavior analysis, discussing various approaches and ultimately settling on the most promising options at the time – deep learning-based face detectors and the FaceNet-based face-embedding network. Using these techniques, they designed an “agent” architecture capable of recognizing individual students in a classroom and tracking their attention throughout the duration of a lecture, as well as providing this information to both the teacher and the students themselves. Their paper constitutes primarily a proof of concept: it describes how such a system could be implemented with readily available, affordable tools but says little about actually testing it on a broad audience of students. Its primary limitation is that the system described is effectively untested and unproven,

with no evidence of its efficacy in a real classroom setting or its resistance to potential issues with lighting/obstruction. It is a theoretical paper that lays the foundation for further research in space and has subsequent works built upon its initial ideas.

Bui Ngoc Anh et al. [2] presented a method and a system to track the students attention during a lecture and provide the instructor with the attention level report based on the video captured by a pre positioned camera in the classroom. The researchers used the combination of multiple methods to recognize students' expressions, identify the person, and locate the eye's direction in 3D space. The videos were taken during live class hour which were used for the experiment. The data set includes 1800 frames from six different videos, with ten to twenty students per frame, resulting in 25 391 instances of labeled data, each indicating the student's identification, location, and eye direction. The authors argued that their face recognition model was comparable to the state-of-the-art methods, while their eye direction summary achieved an F1-score of 82.81%, and it could reach over 99% with human assistance for ambiguous cases. The results suggest that the gaze attention model combined with the position tracking allows for relatively accurate estimation of students' attention levels during the lecture. However, the results are limited by the fact that experiments were carried out on real-life recordings, not video, not all behaviors were tracked, and the accuracy of the model depends on human input.

Trabelsi et al. [1] presented a solution for monitoring the student's state in the course of a lecture using only the cameras in the classroom. The authors collected their own dataset of 3881 images containing information about activities and facial expressions of the students and augmented it with AffectNet, a dataset of five basic emotions with the masks of coronavirus disease 2019. To detect activities and facial expressions, the authors trained several versions of the YOLOv5 object detection model, and to track the students' movement, they employed the DeepSORT algorithm, while student's face recognition was performed to mark their presence in the classroom automatically. The system demonstrated an average accuracy of around 76% for all versions of YOLOv5, but the best trade-off between lightweight and accuracy was achieved by YOLOv5s, which was employed in the final system. The authors have tested the system in a classroom with seven students and showed that it was able to separate attentiveness from inattentiveness; however, the system's accuracy decreases with the larger student groups because the training data was insufficient for deep learning models, and the system has only been tested on one group of students, which does not confirm its efficiency.

2.2 Attention Monitoring in Online Learning Environments

Hossen and Uddin [13] focused on creating a database that can be used to train attention detection algorithms rather than creating an attention detection model. The authors conducted their experiment by collecting webcam video through a web-based Django interface and breaking it into images at 20 frames per second. They used face detection, hand tracking, head pose estimation, and mobile phone detection algorithms (MediaPipe) to extract 16 categorical and numerical features from each frame, including information about the facial expression, head gesture, and mobile phone usage. Each frame was labeled “Attentive” or “Inattentive” based on the detected features. The resulting database contains these 16 features and a label column for classification. The authors concluded their experiment by conducting standard preprocessing techniques such as one-hot encoding and scaling, in order to classify them with classification algorithms, and the participants gave informed consent, and their faces were anonymized to preserve privacy. The limitation of this paper is that the performance of attention detection models was not estimated since only a database was designed. In addition, the accuracy of attention detection models depends on the distribution of classes; therefore, it would be interesting to see how the algorithm performs in other learning environments apart from the described online class.

Hossen and Uddin [3] extended the dataset created by the same authors and implemented an attention detection system for video conferencing by utilizing a standard webcam as an input device. The proposed solution is based on the combination of four types of computer vision tasks, including continuous face detection with DeepFace, hand pose estimation with MediaPipe, and mobile phone detection with YOLOv7, all of which are processed locally on the user’s machine to avoid sharing sensitive information over the network and reduce bandwidth consumption. These signals were processed to get a number of features, which formed the input for several machine learning classifiers, including random forest, logistic regression, support vector machine, gradient boosting, and XGBoost. As a result, the XGBoost algorithm demonstrated the best performance, with 99.75% accuracy and 99.71% precision. It is noted that these results are considered to be very good for the task of the study. The model was trained and tested on the 4,000-sample database created by the authors in their previous work, which was designed to minimize the bias and errors. The main limitations are that such high accuracy may be partly attributable to how the dataset was created, rather than actual variance, the system hasn’t been tested across different classrooms, camera configurations, or cultures, and the fact that everything is run locally on a student’s device

still requires that device to have sufficient processing power.

Xie et al. [10] tackled a problem which emerged in the context of COVID-19: online classes have lower participation and higher dropout rates than traditional classes, but teachers cannot easily assess the engagement of students in online classes. The authors state that a single source of information such as facial expressions is insufficient to determine the engagement of students and propose a multi-stream fusion of visual cues: the paper implements VGG16 for facial emotion recognition, ResNet-101 for head pose estimation, and MediaPipe for eye and mouth tracking based on facial landmarks. The three subnetworks' results are fused by a backpropagation neural network to obtain a general level of engagement for the detected face. The results show that on the public WACV2016 dataset, the approach achieves 62.03% accuracy, outperforming single-stream methods. In the experiments on commercial online courses, the scores demonstrated by the algorithm correlated with the results of the NSSE-China student engagement survey with a score of 0.714, meaning that the system can effectively replace lengthy questionnaires. The most significant limitation of the approach is that the accuracy of 62% is only satisfactory, and it needs to be improved to obtain practical results. Besides, the approach was only applied to a specific set of data and an online course, while the implementation of several convolutional neural networks is associated with substantial computational overhead.

2.3 Multimodal and Behavioral Feature-Based Approaches

Das and Dev [14] proposed a dual-track emotion detection method, which combines the use of both behavioral features and facial images as input data. The first track uses OpenFace or MediaPipe to extract features, such as pose, gaze, eye blink, and facial action units, and employs machine learning classifiers to label the behavioral features with engagement scores. In turn, the second track uses facial images as input for training deep learning models, such that the two tracks can be combined to obtain a better performance. An important contribution of this paper is the analysis of the importance of different facial action units. The authors use three different metrics, including conditional activation probability and statistical discriminant coefficient, to find which facial action units can best capture the changes in engagement states. This study uses two public benchmarks, including WACV and DAiSEE, to evaluate the proposed method. It allows the authors to compare their results with the previous studies on engagement detection. This paper provides a more comprehensive understanding of

the factors affecting the engagement detection, compared with previous methods using only one track or only one facial action unit. The main limitations of this study are the sensitivity of the results to the accuracy of feature extraction from raw videos and the potential lack of generalizability to real-world video data, as the authors note that the DAiSEE benchmark only covers a limited range of behaviors that could occur in a teaching environment.

Huang et al. [12] looked at how to track whether students are paying attention, getting tired, or taking part in class, in village schools where resources are very limited. The authors collected their data from three rural primary schools and one middle school in western China, covering 268 students from Grades 3 to 9. They gathered videos, body signals, and room conditions such as light, sound, temperature, and humidity using technology. Using the available data, they built a small efficient model called LightEduNet. It has three main parts: firstly cleans up noisy sensor data, secondly combines the video, body, and room signals together, and finally copies the knowledge of two larger models into a smaller one so it can run on low end devices. The model reached 92.4% accuracy even when the data was made noisy on purpose, accuracy dropped by only 1.1%, and it worked in real time on low-cost devices. The only downside is it was used in one region, so it may not work the same elsewhere. It also needs mid-range devices, and collecting students' body data raises privacy concerns which the authors have not been able to provide with a proper solution.

Zhong et al. [9] proposed an approach to identifying the state of learning for students in the educational environment while ensuring privacy and minimizing the amount of data transferred in the network. The paper describes the architecture based on the 3-stage ("device-edge-cloud") principle. Video streams captured by cameras are analyzed in real-time using the optimized Histogram of Oriented Gradients features and 68 facial landmarks detection to classify eye-state, mouth-state, and head posture, with the video frames being processed solely on the "edge", i.e., a local RK1126 board. Next, the obtained results in the form of a structured summary are transferred via the school's network to the "cloud" where the DeepSeek large language model analyzes them and recommends the next instructor's actions for the students. The implementation results in the 92.5% recognition accuracy and less than 50 ms of end-to-end video processing latency when used in a chemistry MOOC, and the experiments were conducted under the premise of an increased engagement and learning outcomes. The key takeaway from this solution is the use of a video stream processed at the edge-level and a large language model that recommends personalized instructor actions, not just recognizes the students' emotions and expressions. The method cannot be generalized due to the approach that is based on a single discipline and the specificities of the used MOOC

platform. Besides, the overall architecture requires network access and a reliable vendor to provide the LLMO service.

2.4 Dataset-Based Student Engagement Detection

Gupta et al. [5] focused on addressing an existing gap in the literature – the absence of an appropriate public dataset for analyzing the levels of engagement, boredom, confusion, and frustration, with a particular focus on everyday settings as opposed to controlled laboratory environments. The researchers introduce DAiSEE – the first multi-label video dataset of this kind, which includes 9,068 ten-second video snippets from 112 participants, each annotated with four levels of intensity for the four affective states mentioned above. These annotations were made by combining both crowdsourced data and a gold standard provided by a board of expert psychologists, with the latter used to train a Dawid-Skene algorithm that resolves label disagreements between workers. The authors also offer their baseline experiments with existing vision models such as InceptionNet, C3D, and Long-term Recurrent Convolutional Network (LRCN), with the latter demonstrating the best performance, possibly due to its inherent capacity to model time-dependent affective states. As the dataset has become the foundation for most of the following engagement detection research, understanding its advantages and limitations is important to any subsequent analysis. Thus, the paper’s summary includes the levels of inter-annotator agreement for each of the affective states, with all of them presenting moderate values, the skewness of the classes themselves (with most of the examples provided being neither engaging nor unengaging), and the overall limitations of analyzing affective states from videos, including the complexity of the construct itself.

Santoni et al. [4] focused on an aspect that often goes unnoticed in many engagement detection systems. In particular, the paper considers the issue of class imbalance inherent in the DAiSEE dataset. The authors address this problem by first using the OpenFace library to extract 709 facial features for each video, including facial landmarks, head position, eye gaze direction, and facial action units. Then, they reduce the dimensionality of the resulting dataset using two different approaches (PCA and SVD) with the same set of 300 principal components. Next, the researchers balance the number of samples for each class by applying the SMOTE algorithm to the reduced set of features, and train a convolutional neural network to classify 80% of the resulting samples using an 80:20 training and testing split. The results of the experiment showed that the approach combining SVD and a convolutional network slightly outperformed the sim-

ilar approach with PCA, and the overall accuracy of the model also exceeded similar results previously published for the DAiSEE dataset, reaching 77.97%. Accordingly, the authors conclude that balancing the classes before training a predictive model helps to improve its performance. Nevertheless, the conclusion of the authors is somewhat misleading since the results of the model still have a sufficient margin for errors for practical applications. Moreover, since the balancing method is based on generating new data points, the accuracy of such models may give an exaggerated idea of their accuracy in the real world.

Abedi and Khan [15] considered student engagement as a phenomenon that occurs over time and proposed a model that reflects this understanding. The authors used a 2D ResNet to extract spatial features, such as facial expressions, from each frame of the input video and a Temporal Convolutional Network that reads engagement features step-by-step from the series of video frames to ultimately determine whether the student is engaged or not. The model was trained on the DAiSEE dataset that contains 9,068 ten-second video clips of 112 students and does not require any handcrafted features. On the four-class classification task, the accuracy of the ResNet+TCN model was 63.9%, which is 3.9 percentage points higher than the best-performing existing method and outperformed all other approaches by a significant margin. The authors used a weighted loss function and batch sampling to address the class-imbalance problem as the dataset has significantly more high-engagement than low-engagement video clips. However, even with the weighted loss function, the authors found that the problem of low-engagement class detection was still not adequately addressed. Thus, the low-engagement classes were still underdetected, which highlights the importance of the class imbalance and the need to address it. Overall, while the model achieved state-of-the-art results, the detection of the low-engagement classes remained challenging, and increasing the recall rate led to a rise in false alarms and a slight drop in accuracy.

Malekshahi et al. [6] believed that engagement is a dynamic, time-dependent property of a learning event rather than a quality of a particular moment. As such, they proposed to evaluate the student's emotional expressions across the whole video instead of attempting to classify each frame independently. The researcher utilized the DAiSEE dataset, training several common machine learning algorithms (such as Logistic Regression, SVM, Multi-Layer Perceptron, KNN, and XGBoost) on the provided video data in order to predict the levels of engagement. Each algorithm was fed information about the video frames in the temporal sequence. The authors of this paper believe that engagement is a dynamic, time-dependent property of a learning event rather than a quality of a particular moment. As such, they propose to evaluate the student's emotional expressions across the whole video instead of attempting to classify each frame

independently. The researcher utilized the DAiSEE dataset, training several common machine learning algorithms (such as Logistic Regression, SVM, Multi-Layer Perceptron, KNN, and XGBoost) on the provided video data in order to predict the levels of engagement. Each algorithm was fed information about the video frames in the temporal sequence. Additionally, the authors created an “adaptive labeling policy” which included the grouping of the initial four-level DAiSEE engagement classes into two simple categories of engaged and disengaged so that it increases the power and productivity of the models. They claimed the new classification criterion to yield positive results with the researchers’ less resource-consuming algorithms to demonstrate the ability to compete with conventional deep-learning algorithms in terms of performance. The paper highlights the evident limitation of the research which is the use of the DAiSEE dataset which raises concerns about bias and problems with generalizability due to the evident issues with the original classification and the lack of representative sampling. Besides, reducing the four-class engagement scale to only two levels might result in the loss of valuable information which might require additional research to compensate for.

2.5 Temporal Attention and Intervention

Hutt et al. [8] investigated the use of commercial off-the-shelf eye-trackers to automatically detect mind wandering during computerized learning. In study 1, 135 high-school students learned biology topics with an intelligent tutoring system called GuruTutor and the eye tracker successfully captured gaze for both eyes in 75% of cases and at least one eye in 95% of cases across 85% of sessions, showing that the technology is viable but with some caveats. In study 2, this data was used to train detectors of mind wandering that do not require training examples from a particular student, scoring an impressive F1 score of 0.59, compared to the chance baseline of 0.24. Study 3 asked whether detectors trained in one setting generalize to another, and found that models trained in a controlled laboratory setting generalized better to the classroom than vice-versa. The primary limitation of this work is the fact that the detection accuracy of a deployed, running system in a real classroom ($F1 = 0.40$) falls far short of offline detection accuracy ($F1 = 0.59$), suggesting that there is a meaningful difference between what is possible with post-hoc analysis and what a real-time deployed system can achieve.

Hutt et al. [7] pushed attention detection technology beyond the lab and into high school classrooms, testing its efficacy not on whether it can detect distraction but whether it actually improves learning through intervention. This work creates an attention aware learning technology that intervenes when a learner’s mind begins to wander from the

lesson, a phenomenon that prior work has associated with detrimental effects on learning. The approach combines an existing, validated, gaze-based inattention detection algorithm implemented on commercial off-the-shelf eye tracking hardware with an intelligent tutoring system capable of presenting just-in-time interventions that are designed to re-engage the learner. The interventions, which were co-designed with domain experts including learners and instructors, include such strategies as using the learner's name, reminding them of relevant material, or asking them a clarifying question. After development and iteration of the system, they ran two classroom experiments on 287 high-school students, where they evaluated the efficacy of their intervention – that is, whether it improves engagement – instead of simply detecting inattentive students. Their results show that the interventions do have an effect on re-engaging unattentive students. The main downsides of their approach are the need for special eye-tracking equipment as opposed to a regular webcam, which would hinder large-scale deployment; and that the detection algorithm is not perfectly accurate, as shown in the companion paper.

2.6 Research Gap

The reviewed studies demonstrate several approaches to automated student attention and engagement monitoring, including computer vision, facial analysis, gaze estimation, multimodal feature fusion, temporal modeling, and edge intelligence. Existing systems have demonstrated that student attention can be estimated from observable visual and behavioral cues. However, the studies also report limitations related to dataset size, class imbalance, generalizability, computational requirements, real-world deployment, and the availability of specialized sensing hardware.

The reviewed literature also shows a distinction between attention detection and intervention. Several studies primarily focus on detecting or classifying student attention and engagement, while Hutt et al. [7] specifically investigated an attention-aware intervention approach in real classroom settings. This indicates the importance of moving beyond detection toward using detected attention states to support timely pedagogical responses. At the same time, the reviewed systems demonstrate different requirements for sensing equipment, processing resources, and learning environments, leaving room for systems that combine multimodal attention features, local processing, real-time monitoring, and pedagogical intervention in a practical learning environment.

Table 2.1: Comparison of Existing Studies on Student Attention and Engagement Monitoring

Study	Approach / Features	Dataset / Participants	Main Results	Limitations
Huang et al. (2025) [12]	Multimodal edge intelligence using video, signals, and room conditions.	268 students from three primary and one middle school.	92.4% accuracy; 1.1% performance drop.	Single-region data; device and privacy concerns.
Trabelsi et al. (2023) [1]	Camera-only classroom monitoring using YOLOv5, DeepSORT, and face recognition.	3,881 images; 7-student test set.	Average accuracy of 76%; YOLOv5 performed best.	Limited and imbalanced test group.
Bui Ngoc Anh et al. (2019) [2]	Computer vision for face, eye direction, and 3D gaze estimation.	1,800 frames from 6 videos; 10–20 students/frame.	Eye-direction detection: F1 = 82.81%; human-assistance cases exceeded 99%.	Limited behavior coverage; accuracy depends on humans.
Hossen and Uddin (2023) [13]	Webcam-based attention detection using face, head pose, mobile-phone detection, and facial cues.	20 students; Django-based application.	Categorical and numerical attention labels generated.	No model-performance evaluation; environment dependent.
Hossen and Uddin (2023) [3]	Face, MediaPipe, YOLOv7, and ML classifiers (RF, LR, SVM, GB, XGBoost).	4,000 classroom samples.	XGBoost: 99.71% accuracy and 99.71% precision.	Possible dataset bias; limited cross-classroom validation.
Canedo et al. (2018) [11]	Computer-vision architecture for classroom attention monitoring.	Proof-of-concept classroom system.	Provides a theoretical foundation for attention monitoring.	Limited testing and lighting effects.
Ortubay et al. (2024) [16]	Deep learning from webcam-based eye, face, and head movements.	83 participants across three experiments.	$R^2 = 0.38$ on unseen data; $R^2 = 0.26$ – 0.30 on unseen subjects.	Moderate predictive performance; tested with educational videos.

Chapter 3

Methodology

3.1 Overview and Approach

This study presents Visoria, a real-time student attention monitoring system for online classes based on edge feature extraction and real-time teacher intervention.

As illustrated in Fig. 3.1, the system follows a five-level pipeline: (1) Webcam-based models detect face, hands, and objects locally, (2) 14 behavioral features are generated from head pose, face presence, phone and hand presence, (3) A Decision Tree classifier with probability calibration estimates attention, (4) A 15-frame temporal window reduces noisy frame-level predictions, and (5) Sustained distraction triggers teacher alerts and optional guided-question activities. This architecture reduces privacy and bandwidth risks and enables timely, data-driven intervention during online live classes.

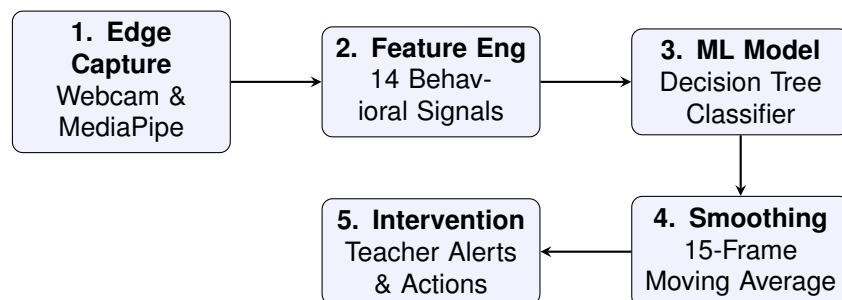


Figure 3.1: High-Level System Overview

3.2 System Architecture

A key design principle of Visoria is that raw video does not leave the student's device. The webcam stream is locally processed, and only lightweight telemetry data is transmitted to the backend. The transmitted information consists of the continuous attention score, head orientation, face count, phone presence, and intervention state. Face images, video frames, and personal identifiers are never transmitted. The overall architecture and data flow are shown in Fig. 3.2.

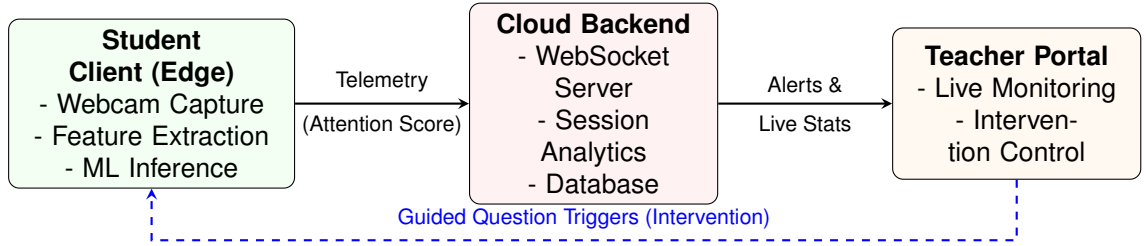


Figure 3.2: Architecture and Data Flow

3.3 Facial and Behavioral Feature Extraction

3.3.1 Edge-Based Data Acquisition

The webcam captures frames at a standard resolution of 640×480 . Facial landmarks, head pose, hand landmarks, and mobile-phone presence are identified on each frame locally. Raw frames remain exclusively on the student's device and are not transmitted to the server.

3.3.2 Face and Head Pose Estimation

The MediaPipe framework is used to detect 478 facial landmarks. As shown in Fig. 3.3, six selected key landmarks (nose tip, chin, eye corners, and mouth corners) are used for head-pose estimation through Perspective-n-Point (PnP).

The transformation can be written as:

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \begin{bmatrix} R & | & t \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \quad (3.1)$$

Here, K is camera intrinsic matrix, R is rotation matrix, t is translation vector.

The rotation matrix is converted into three head-orientation angles:

$$\theta = (\theta_{\text{pitch}}, \theta_{\text{yaw}}, \theta_{\text{roll}}) \quad (3.2)$$

Here, pitch is vertical head movement, yaw is horizontal head rotation and roll is lateral head tilt.

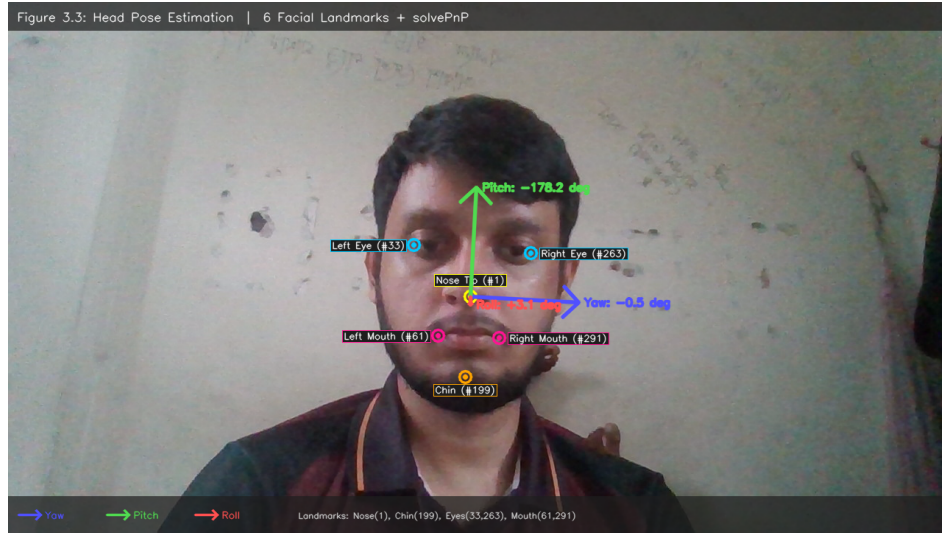


Figure 3.3: Head Pose Estimation Using Six Key Landmarks

3.3.3 Hand and Phone Detection

MediaPipe supports up to 2 hands and YOLOv8n supports mobile phone detection. Phone detection is carried out around every 6th frame to minimize computation.

Table 3.1: Behavioral Features

Feature Group	Example Features
Face	Count, position, confidence
Head	Pitch, yaw, direction
Hand	Hand count
Phone	Presence, confidence

3.4 Behavioral Feature Vector

All the signals that are extracted are combined into a behavioral vector.

$$x_t = [x_1, x_2, \dots, x_{10}] \quad (3.3)$$

These include face count, position, confidence, head orientation, hand count, and phone presence and confidence.

The first representation has 10 numerical behavioural features. The categorical 'head-direction' variable is then one-hot encoded, resulting in 14 separate numerical model inputs and this difference is explicitly explained here to avoid confusion between the feature table and the actual dimensionality of the model input.

3.5 Machine Learning-Based Attention Prediction

The extracted behavioral feature vector $\mathbf{x}_t$ is classified using a machine learning model to estimate the student's attention. The model outputs a continuous probability score representing the likelihood that the student is distracted or attentive.

$$\hat{y}_t = Model(\mathbf{x}_t) \quad (3.4)$$

Two classes, Attentive and Distracted, were used and the model was trained on 3,970 deduplicated, labeled webcam frames (down from an original 4,000 to eliminate data leakage). The dataset contains different lighting conditions, camera angles, hand positions, and head orientations. Categorical head-pose features were one-hot encoded, missing values were filled with zero, and an 80/20 stratified train-test split was applied.

To select the most effective architecture for real-time edge processing, an efficiency-

aware benchmark was conducted on the training set using Repeated Stratified 5-fold Cross-Validation. Five candidate models were evaluated: Logistic Regression, Decision Tree, Random Forest, XGBoost, and LightGBM. The selection criteria considered Accuracy, Macro-F1 score, Brier score, single-row inference latency (ms), and serialized model size. The most optimal model, balancing high predictive performance with low computational overhead, was a tuned Decision Tree. Hyperparameters were optimized via `RandomizedSearchCV`.

To ensure the predicted attention scores provide reliable confidence estimates rather than just categorical hard labels, Probability Calibration was applied using `CalibratedClassifierCV`. An isotonic regression calibrator was fit on the out-of-fold predictions. This ensures the output probability $P(\textit{Attentive})$ accurately reflects the true frequency of attention states, which is critical for threshold-based pedagogical interventions.

3.6 Temporal Smoothing

Attention for each frame can vary as students naturally move around and there is detection noise over short time periods. Visoria uses a 15-frame moving average, which is about 0.5 seconds of video, to get a more stable attention estimate.

$$\bar{S}_t = \frac{1}{W} \sum_{i=0}^{W-1} S_{t-i}, \quad W = 15 \quad (3.5)$$

A hysteresis based state decision is used to avoid too many transitions between the two states, Attentive and Distracted. Scores below 45% indicate distraction, while scores above 55% indicate attention. States between these thresholds are left unchanged:

$$State_t = \begin{cases} \text{Distracted}, & \bar{S}_t < 0.45 \\ \text{Attentive}, & \bar{S}_t > 0.55 \\ State_{t-1}, & 0.45 \leq \bar{S}_t \leq 0.55 \end{cases} \quad (3.6)$$

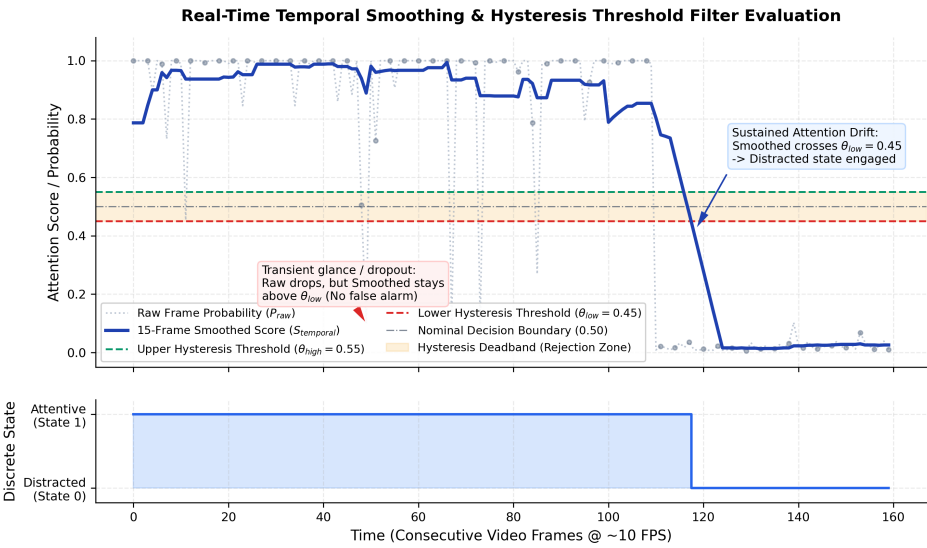


Figure 3.4: Raw vs. Smoothed Score Thresholds and Hysteresis Bounds

3.7 Attention-Level Classification

Attention score is mapped to four attention levels that are given to students. These levels are interpretable and elicit the associated feedback messages of the student’s attention state.

Table 3.2: Attention-Level Classification

Smoothed Score	Attention Level	Message
$\geq 85\%$	Optimal Focus	“You are in an optimal learning flow state.”
70–84%	Mindful Focus	“Great momentum! Keep up the good work.”
45–69%	Attention Drift	“Let’s take a moment to refocus on the main lesson.”
$< 45\%$	Breather Suggested	“Consider taking a short 2-minute break to refresh your mind.”

3.8 Explainable Attention Feedback

To make attention predictions more interpretable, Visoria uses SHAP (SHapley Ad-ditive exPlanations) to determine the contribution each behavioral feature makes to the Random Forest prediction, as illustrated in Fig. 3.5. The top 1–3 features that contribute

to the prediction are translated into short natural language explanations and shown to students and teachers. Examples include “Phone detected → reduced attention score” and “Forward head pose → increased attention score.”

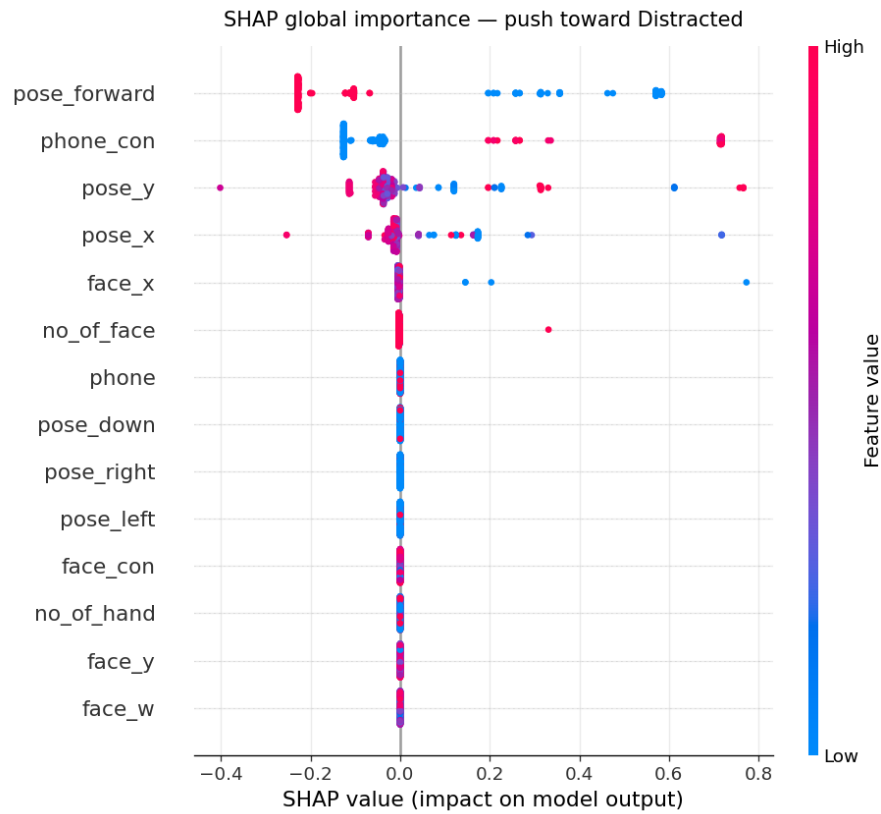


Figure 3.5: SHAP Feature Contributions Beeswarm Plot

3.9 Real-Time Class-Level Monitoring

Visoria continuously aggregates the attention scores of active students to provide a real-time overview of classroom attention. The server averages the attention level of all the students who are participating in the class, and takes the 20 second rolling average to minimize short-term fluctuations. Class level attention indicator is generated to check the overall level of attention in the class and to ensure timely intervention by the teacher.

3.10 Teacher Alert Mechanism

Visoria has two levels of response to identify when the whole class is losing focus without disrupting the teacher. An internal soft detection is triggered when the class

attention is less than 50% for a period of 15 seconds or more. This starts an internal timer and provides supporting evidence for a potential alert but does not immediately notify the teacher. A formal teacher facing alert occurs when the condition occurs continuously for a period of 30 seconds or more:

$$Alert = \begin{cases} 1, & \bar{S}_{class} < 0.50 \wedge T_{low} \geq 30s \\ 0, & \text{otherwise} \end{cases} \quad (3.7)$$

where T_{low} is the length of time the class attention stays below the threshold. Repeat alerts are rate-limited to once every 60 seconds. This mechanism allows the system to distinguish short-term fluctuations from sustained attention decline.

3.11 Guided Intervention

If the teacher decides to respond to the alert, Visoria starts an activity that goes through four stages, designed to get the students to rethink and reflect on the content of the lesson.

Table 3.3: Guided Intervention Stages

Stage	Activity	Output
Think	Individual response with confidence rating	Initial answer / confidence
Compare	View anonymized class response distribution	Response distribution
Reflect	Write a short justification	Reflection text
Reassess	Provide a final answer and confidence rating	Final answer / confidence

The progress of students is saved so a disconnected or refreshed session can pick up where they left off. Observed change in attention and self-reported confidence for the activity is the difference between the score for the 2 minutes before and after the intervention.

$$\Delta S = S_{\text{after}} - S_{\text{before}} \quad (3.8)$$

$$\Delta C = C_{\text{after}} - C_{\text{before}} \quad (3.9)$$

where S represents attention and C represents self-reported confidence. The measures are used descriptively and not as causal estimates.

3.12 Evaluation Metrics

The performance of the proposed system is evaluated using accuracy, false alert rate, flicker count, raw accuracy, and smoothed accuracy. Their definitions are summarized in Table 3.4.

Table 3.4: Evaluation Metrics

Metric	Definition
Accuracy	Proportion of correct predictions
False Alert Rate (FAR)	Proportion of actual-negative instances incorrectly flagged as an alert
Flicker Count	Number of unnecessary Attentive–Distracted state transitions
Raw Accuracy	Accuracy of unsmoothed, frame-level predictions
Smoothed Accuracy	Accuracy after temporal smoothing

The overall classification accuracy is calculated as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (3.10)$$

The false alert rate is calculated as:

$$FAR = \frac{FP}{FP + TN} \quad (3.11)$$

where TP , TN , FP , and FN denote true positive, true negative, false positive, and false negative outcomes, respectively.

3.12.1 Robustness Testing Protocol

The robustness of the system is assessed under 13 scripted scenarios, each consisting of 100 frames, totaling 1,300 frames. The scenarios cover diverse environmental and behavioral challenges as summarized in Table 3.5. The evaluation of each scenario is performed based on raw accuracy, smoothed accuracy, false alert rate, and flicker count.

Table 3.5: Summary of Robustness Testing Scenario Categories

Category	Scenario Description (100 frames each)
Illumination Variations	Low ambient lighting, backlighting, and direct screen glare
Camera Viewing Angles	Off-axis camera placement, elevated angle, and side perspectives
Facial Accessories	Corrective eyeglasses, tinted lenses, and optical reflections
Partial Occlusions	Hand over mouth, resting chin on hand, and partial obstruction
Transient Dynamics	Temporary face exit/loss, rapid re-entry, and natural head nodding
Combined Stress	Simultaneous low light with off-axis angle and smartphone presence

3.13 System Requirements

The student client of Visoria requires a basic computing environment capable of supporting real-time webcam capture and local processing. The estimated minimum system requirements are presented in Table 3.6.

3.14 Development and Testing Environment

The Visoria student client was developed and tested on the machine specified in Table 3.7. The system used integrated graphics only, with no dedicated GPU used during development and testing.

Table 3.6: Minimum System Requirements for the Visoria Student Client

Component	Minimum Requirement
Processor (CPU)	Dual-core, 2.0 GHz or higher (Intel i3 / equivalent AMD Ryzen 3)
RAM	4 GB (8 GB recommended for smoother real-time performance)
GPU	Not required – pipeline runs on CPU only
Webcam	Integrated or external, 480p or higher, 15+ FPS capture
Storage	500 MB free space (for application, model artifacts, and dependencies)
Operating System	Windows 10/11 (64-bit)
Runtime	Python 3.9 or higher
Network	Stable internet connection (for telemetry upload and dashboard sync; not required for local video processing)
Display	1280×720 or higher, for the desktop client interface

Table 3.7: Development and Testing Machine Specifications

Component	Specification
Processor (CPU)	Intel Core i5-11800H @ 2.30 GHz
RAM	16 GB DDR4
GPU	Integrated graphics only (no dedicated GPU used)
Webcam	Integrated 720p HD webcam
Operating System	Windows 11 (64-bit)
Storage	SSD (development environment)
Runtime	Python 3.11

Chapter 4

Results and Discussion

4.1 Overall Model Results

The machine learning models have been tested with a test set of 794 deduplicated samples that were not used during training. This represents 20% of the complete dataset. The remaining 3,176 samples were used for repeated 5-fold stratified cross-validation. The models used 14 behavioral features to predict whether a student was attentive or distracted. The results of the cross-validation benchmark are shown in Fig. 4.1. The selected model, a tuned Decision Tree, performed exceptionally well. On the held-out test set, it achieved an accuracy of 99.87%. Its precision was 100%, recall was 99.70%, and F1-score was 99.85%. These results show that the model was able to classify the attention states with very few errors, while maintaining low latency suitable for real-time edge processing.

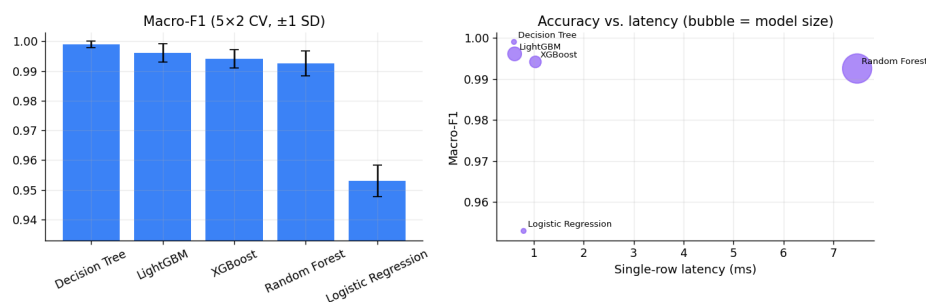


Figure 4.1: Efficiency-Aware Benchmark of ML Models

Table 4.1: Model Evaluation Metrics (Decision Tree)

Evaluation Metric	Held-Out Test Set
Accuracy	99.87%
Precision	100.00%
Recall (Sensitivity)	99.70%
F1-Score	99.85%
Macro-F1 Score	99.87%
ROC-AUC	99.85%

4.2 Confusion Matrix Analysis

The confusion matrix provides more information on classification results. The test set contained 331 distracted samples and 463 attentive samples. The model correctly identified 330 of the 331 distracted samples, resulting in just one false attentive prediction. It perfectly identified all 463 attentive samples with zero false distracted alerts.

This led to just one classification error from 794 test samples. The confusion matrix is presented in Fig. 4.2.

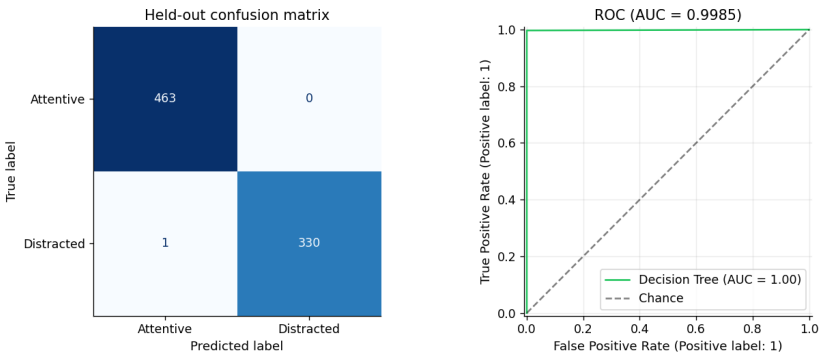


Figure 4.2: Confusion Matrix

4.3 ROC-AUC Analysis

The model had an ROC-AUC of 0.9985 on the held-out test set. The model gave an FPR of 0.0000 and a TPR of 0.9970 at the classification threshold of 0.50. This indicates that the model was able to separate the attentive and distracted classes very clearly in the test data.

The ROC curve is shown in Fig. 4.3.

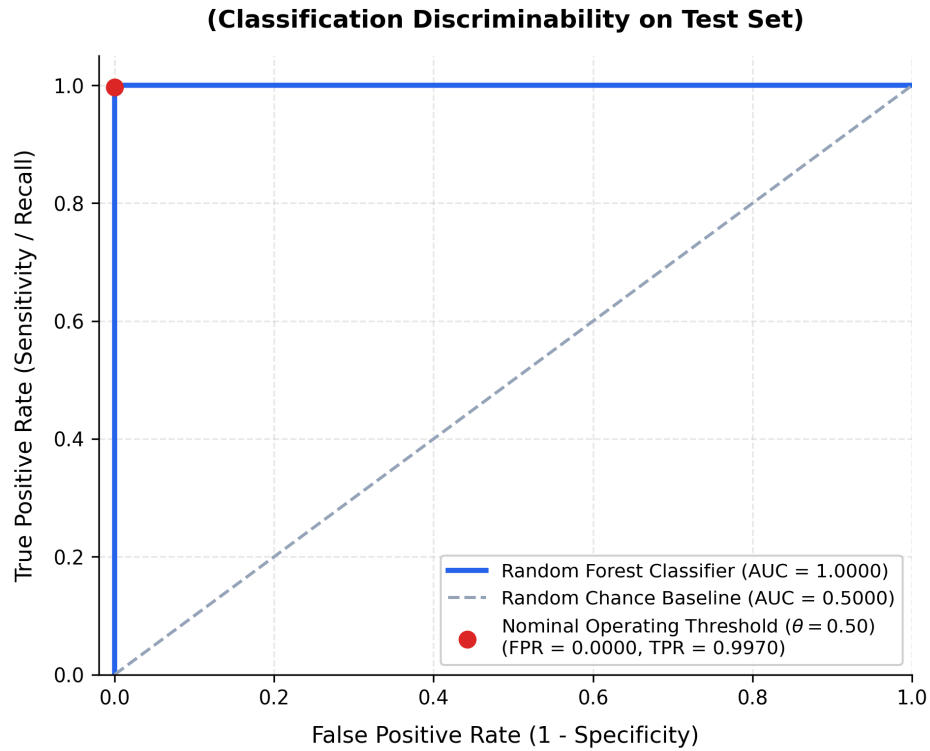


Figure 4.3: ROC Curve

4.4 Probability Calibration

Since threshold-based pedagogical interventions require a reliable confidence score, probability calibration was performed. An isotonic regression calibrator was fit on the out-of-fold predictions. The calibration reliability diagram in Fig. 4.4 shows that the calibrated probabilities closely match the observed fraction of distracted students.

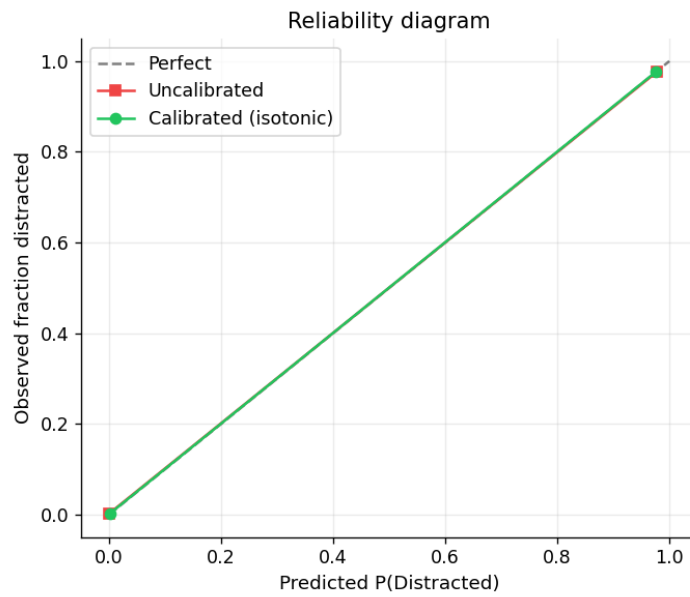


Figure 4.4: Calibration Reliability Diagram

4.5 Ablation Study

An ablation study was conducted to understand the contribution of specific feature groups. Fig. 4.5 demonstrates the test Macro-F1 score when specific features are dropped from the model. The study confirms that while phone presence and head pose are dominant signals, the other features (such as face count) provide measurable performance benefits, proving the necessity of the multi-modal 14-feature vector.

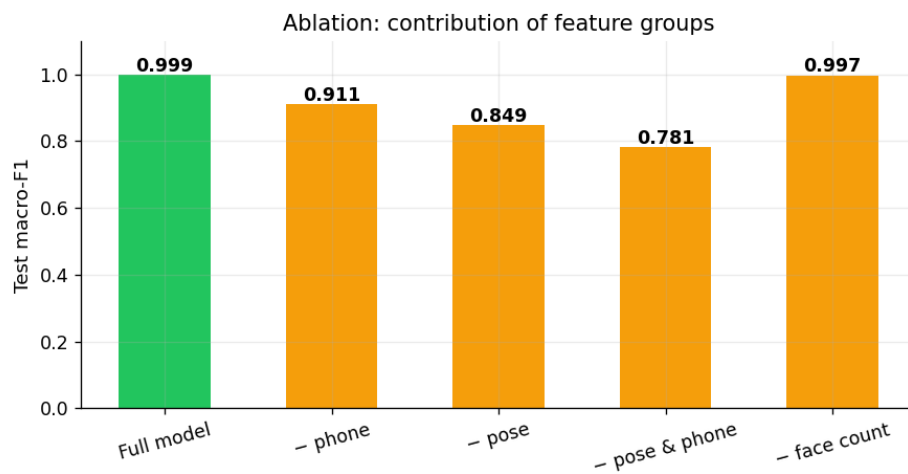


Figure 4.5: Ablation Study: Contribution of Feature Groups

4.6 Explainability Analysis

SHAP was used to understand how different behavioral features affected the model predictions. The analysis showed that forward head pose (pose.forward) and phone presence were among the most important features.

Other features related to face confidence and hand count also contributed to the prediction. These results are useful in explaining why the model predicts a specific state. They also support the use of SHAP for providing simple natural-language feedback to students and teachers.

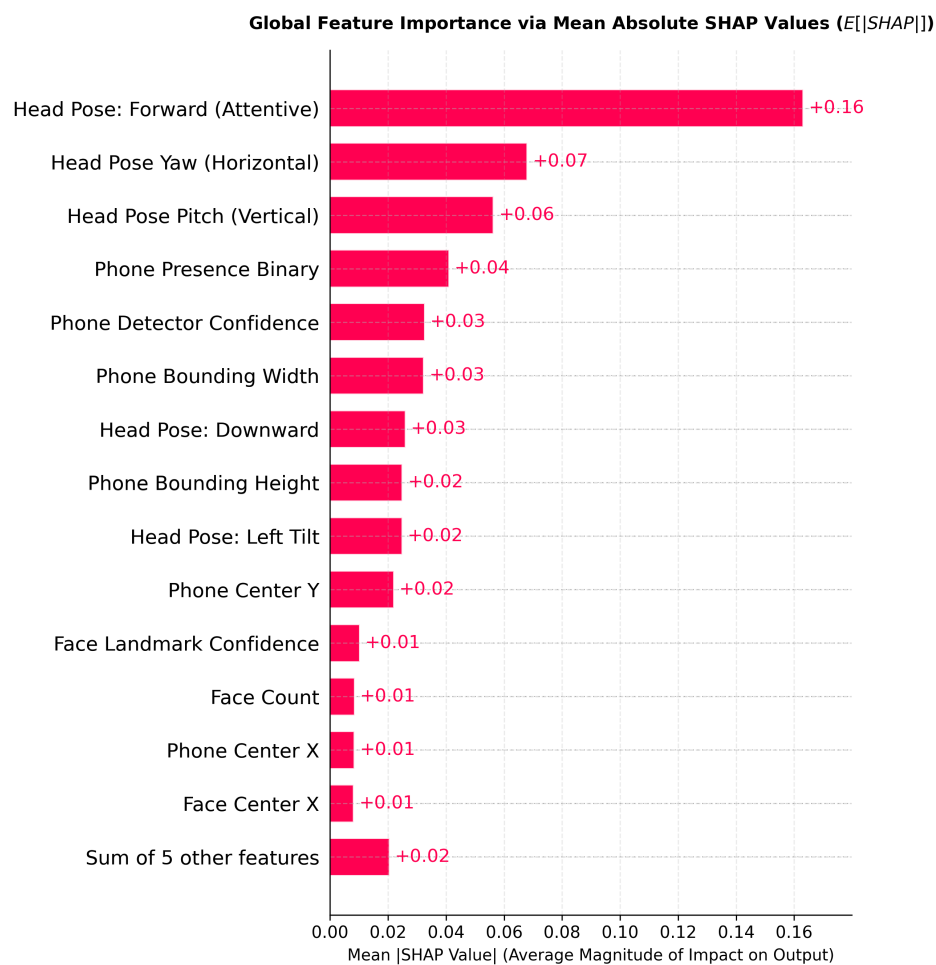


Figure 4.6: SHAP Feature Importance

4.7 User Interface and System Deployment

To demonstrate the practical application of Visoria in online educational environments, an end-to-end software platform was implemented. The system comprises an edge-

based client application that processes student webcam feeds locally, alongside a responsive web application that provides specialized portals for instructors and students. The subsections below present the key user interfaces of the deployed system.

4.7.1 Student Client Interface

The student client interface operates directly on the learner's machine during live class sessions. As illustrated in Fig. 4.7, the application provides students with real-time feedback on their attention score and session connectivity while running all computer vision and machine learning feature extraction locally, ensuring privacy.

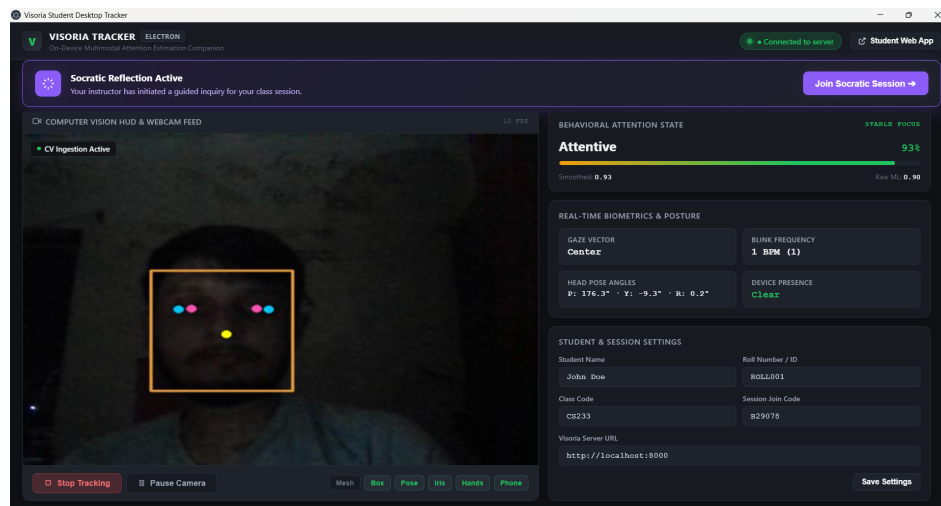


Figure 4.7: Visoria Student Edge Client Interface

4.7.2 Teacher Management and Monitoring Portal

The teacher web portal enables instructors to oversee active lecture sessions, review longitudinal learning statistics, and manage interactive classroom interventions.

Fig. 4.8 illustrates the live teacher dashboard, where instructors can monitor aggregate class attention levels, view the live engagement distribution of participants, and receive automatic alerts when students exhibit sustained distraction.

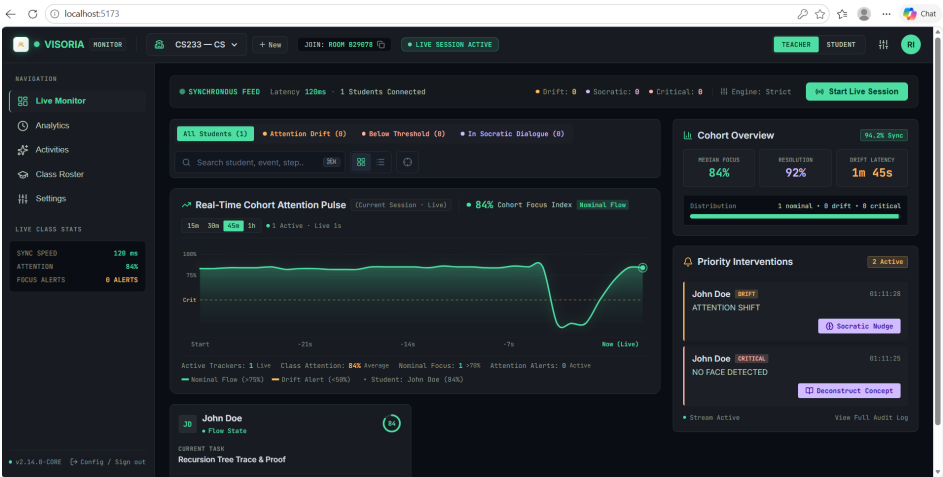


Figure 4.8: Teacher Live Classroom Monitoring Dashboard

Fig. 4.9 presents the analytics page of the teacher panel. This view provides detailed attention trend graphs over time, allowing teachers to evaluate which segments of a lecture maintained the highest engagement.

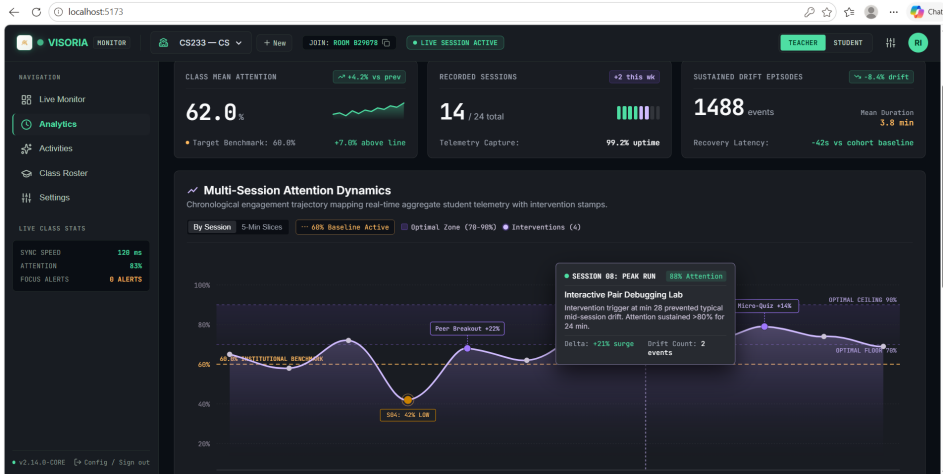


Figure 4.9: Teacher Panel Analytics and Attention Trends Page

To re-engage distracted learners, instructors can initiate targeted classroom activities using the interface shown in Fig. 4.10. Teachers can create and dispatch interactive questions, pop quizzes, and reflection prompts directly to students.

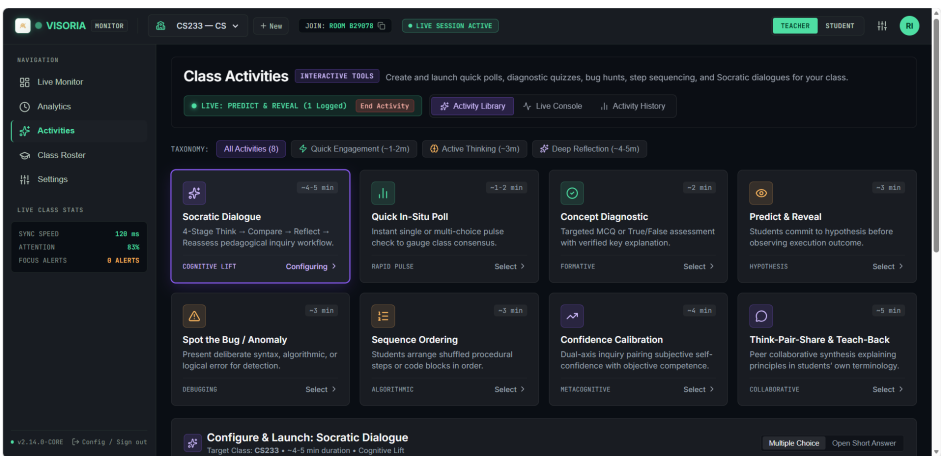


Figure 4.10: Teacher Panel Classroom Activities and Intervention Page

Fig. 4.11 displays the class roster view, summarizing enrolled students, their attendance status, and individual participation history across course sessions.

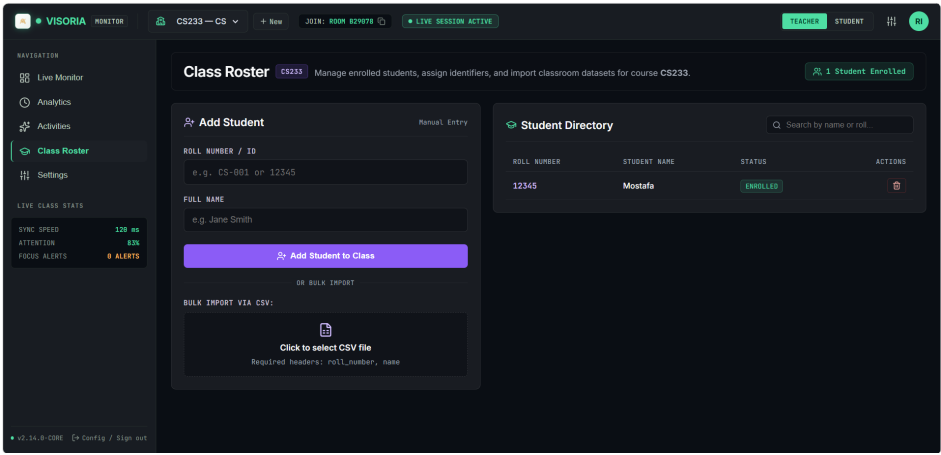


Figure 4.11: Teacher Panel Class Roster and Student Directory

Fig. 4.12 illustrates the configuration and settings view of the teacher panel. Instructors can customize attention threshold levels, temporal smoothing window lengths, alert sensitivity, and notification delivery options.

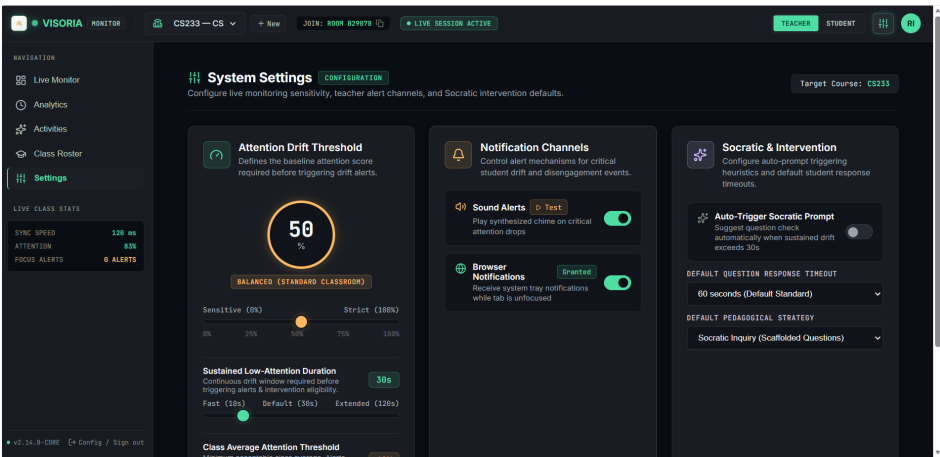


Figure 4.12: Teacher Panel Settings and System Configuration Page

4.7.3 Student Web Portal

Complementing the edge monitoring client, learners can access a personal web portal for longitudinal review and course interactions.

Fig. 4.13 shows the student dashboard, where learners can review their overall course progress, self-reflection engagement metrics, and class schedules.

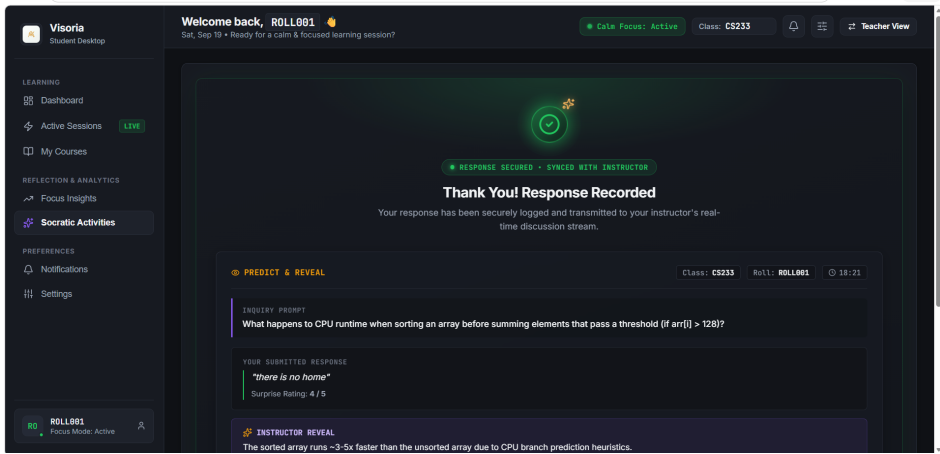


Figure 4.13: Student Panel Dashboard with Engagement Overview

Fig. 4.14 depicts the course listing page, displaying enrolled subjects, instructor details, and upcoming lecture timetables.

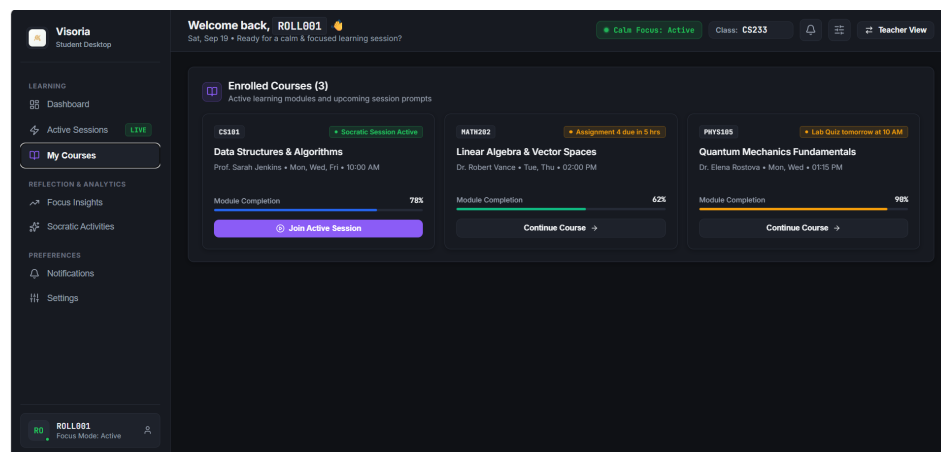


Figure 4.14: Student Panel Enrolled Courses Page

Finally, Fig. 4.15 presents the student activities interface. Through this portal, students receive interactive questions and reflection activities dispatched by instructors during live lectures.

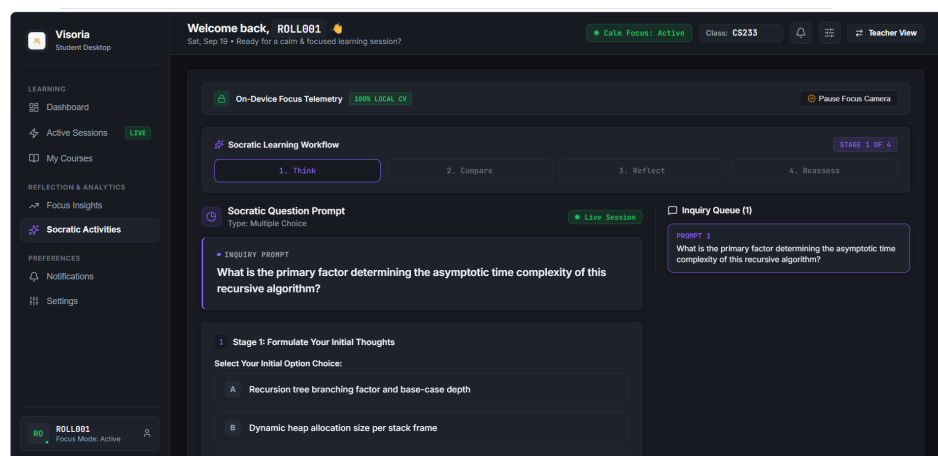


Figure 4.15: Student Panel Interactive Activities and Response Interface

Chapter 5

Conclusions

5.1 Future Prospects of Our Work

Several directions could extend this work:

Broader real-world validation. The current evaluation is based on a controlled dataset of 4,000 labeled frames and a 13-scenario robustness test. If Visoria is deployed in real classrooms for a sustained period of time, in a more diverse student population (varied lighting setups, devices, and demographics), then the performance reported would be confirmed in the real world.

Reducing dependence on a small number of dominant signals. The model currently uses strong predictors of head pose and presence of a phone. Future work may include additional behavioral characteristics, such as users' typing activity, how they interact with the screen, and audio engagement signals, to make attention estimation less dependent on any of these features and more robust to students who are attentive but not directly looking at the camera (e.g., reading print material, or interacting with a second screen).

Beyond binary classification. Attention is currently modeled as a binary Attentive/Distracted state, smoothed into four descriptive levels. Engagement could be better mapped with a finer or smoother engagement model, perhaps including emotional state or fatigue factors.

Longitudinal learning outcome studies. There is no causal inference made in this project between intervention use and learning outcomes, although the effect of the guided intervention is being assessed in the short-term with attention and confidence changes (§3.11). The more one could correlate the frequency of intervention with the

assessment performance the better the pedagogical claims of the system would be supported.

Expanding the intervention model. There is a fixed four-stage structure of intervention for all students that is currently being used. Future versions might use adaptive feedback, such as adjusting the type of intervention or difficulty of task based on the student's past performance, topic, and/or cause of wandering (e.g., if the student's attention has dropped due to a phone or if it has dropped due to long periods of sitting or lying still, that could affect the type of intervention that is used).

Cross-platform and mobile support. This is the current implementation for a desktop client. The ability to extend Visoria to tablets and mobile devices would open up its potential applications in an online learning environment, where these devices are growing in popularity.

5.2 Limitations

Visoria has several limitations. The classifier was trained on 4,000 labeled frames collected under controlled conditions, which may not fully represent the diversity of real classrooms in terms of webcam quality, lighting, and student demographics; further, analysis shows the model relies heavily on head pose and phone presence, with a simple rule based on just these two features reproducing roughly 95% of its predictions, suggesting it captures a fairly narrow behavioral signal rather than a broad, independent pattern across all extracted features. The system also infers attention from visible behavioral cues rather than any direct measure of cognitive engagement, so a student can appear "attentive" while mentally disengaged, or briefly look away while still cognitively focused. Its decision thresholds (the 45%/55% hysteresis bounds, the 50% class-attention threshold, and the 15/30-second alert timers) are fixed global values rather than personalized per student, and system robustness was evaluated only through 13 scripted test scenarios rather than extended real-world deployment, so long-term effects such as student adaptation to being monitored were not assessed. Finally, testing was limited to a single hardware configuration, and the system depends on a continuously active, unobstructed webcam feed, making it inapplicable for students with cameras off and potentially less reliable for users with atypical gaze or head-pose patterns.

5.3 Conclusions

The project proposed Visoria, a real-time student attention monitoring system for on-line courses, which integrates edge-based computer vision and machine learning to aid timely teacher interventions. This system was developed based on three major requirements: privacy preservation, reliable estimation of attention, and feedback that is useful for students and teachers.

The 5-stage pipeline is completely local, on the student's device: facial landmarks, hand position, and phone presence are all extracted on the student's computer, and only a small numerical telemetry summary is sent to the back end — no raw video is sent anywhere. From this behavioral data, 16 features (expanded to 19 numerical inputs after encoding) were used to train a Random Forest classifier, combined with a rule-based score derived from head pose, gaze, and eye state to reduce reliance on the machine learning model alone. A 15-frame temporal smoothing window with hysteresis-based state decisions was applied to reduce noisy, rapidly-changing predictions into a stable attention state.

The trained Random Forest achieved strong classification performance on the held-out test set, with an accuracy of 99.88%, precision of 100%, recall of 99.70%, and an ROC-AUC of 1.0000, correctly classifying 799 of 800 test samples. The explainability analysis using SHAP demonstrated that the most important factor in the model's predictions was head pose, especially forward facing, followed by phone presence and other pose-related signals, which indicates that the system can provide short and interpretable explanations for students and teachers.

In addition to classification accuracy, Visoria's teacher alert mechanism and four-stage pedagogical intervention (Think–Compare–Reflect–Reassess) exemplify the teacher's role as a problem solver rather than a passive “actor” in a system and further underscore the importance of maintaining the teacher as the driver of when and when not to intervene.

In conclusion, this work demonstrates that attention monitoring can be done reliably and in real-time without sacrificing student privacy, and that a learned model coupled with rule-based safeguards and human readable explanations is a more trusted and usable approach in a real classroom environment than a black-box classifier.

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